

Procedure: CNS Experiment-2

Aim: Implementation of an AAA server as a user authentication and authorization technique for remote access to a network device using Cisco Packet Tracer.

Step-by-Step Procedure

Step 1: Create Network Topology

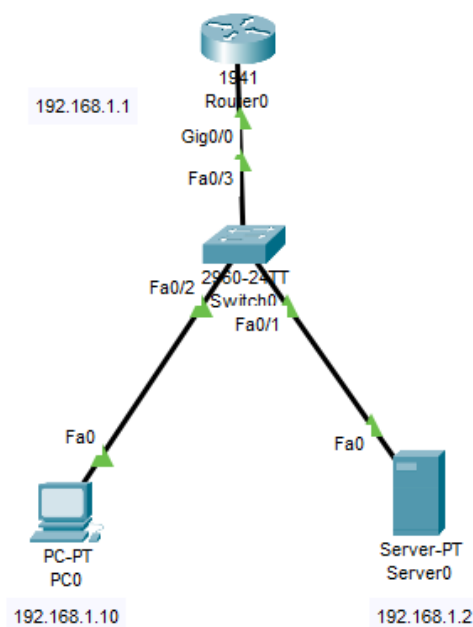


Fig.1. Topology

Step 2: Configure IP Addresses

- **Server Configuration**

Set:

IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

- **PC Configuration**

Set:

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

- **Router Configuration**

Use CLI Commands:

```
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

Or use Config tab:

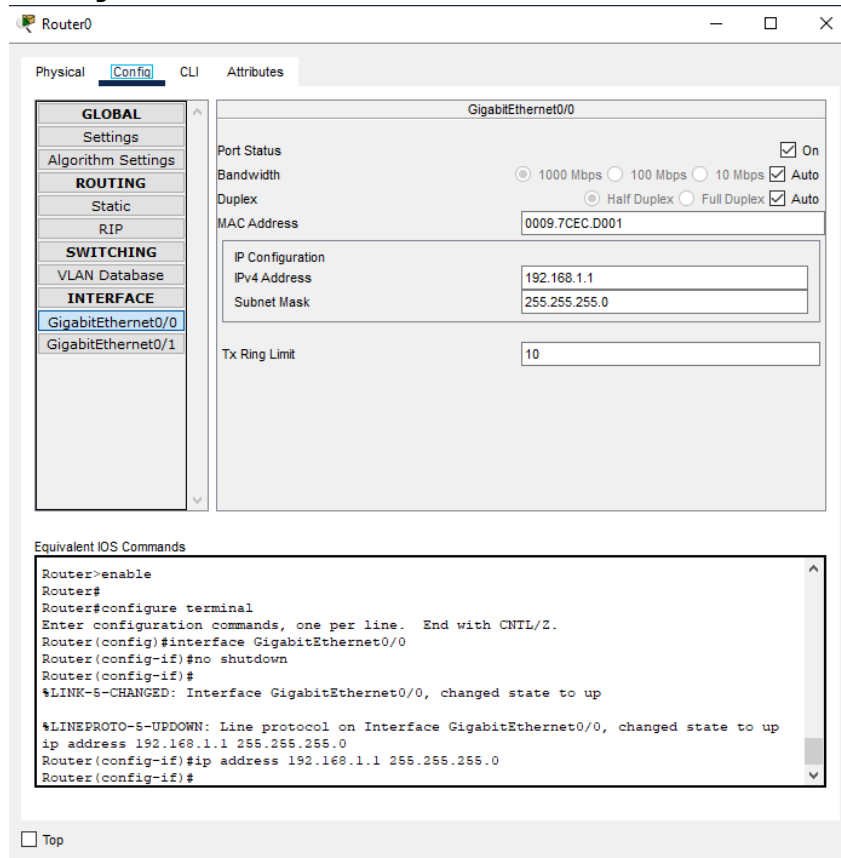


Fig.2. Router interface

Step 3: Configure AAA Server

1. Click Server → Services → AAA
2. Turn AAA service ON
3. Set the following:

Add Router as Client: AAA server must trust the router before processing authentication requests.

Client Name: R1

Client IP: 192.168.1.1

Secret: cisco

ServerType: Radius

Add

Add User Credentials: These credentials will be used by remote users to log in.

Username: admin

Password: cisco123

Add

Step 4: Enable AAA on Router

On Router CLI:

```
enable
```

```
configure terminal
```

```
aaa new-model
```

- This command activates AAA functionality on Cisco router. Without this, AAA commands will not work.

- Router switches from simple authentication to centralized AAA control

Step 5: Configure Router to Use AAA Server

If using RADIUS:

```
radius-server host 192.168.1.2 key cisco
```

If using TACACS+:

```
tacacs-server host 192.168.1.2 key cisco
```

What it does: It tells router where the AAA server is located and specifies shared secret for secure communication.

Step 6: Configure Authentication Method

```
aaa authentication login default group radius local
```

OR for TACACS+:

```
aaa authentication login default group tacacs+ local
```

When a user logs in: Router checks AAA server first, if server unavailable → checks local database. It provides reliability and centralized control.

Step 7: Configure Local Backup User

```
username backup privilege 15 secret backup123
```

Purpose:

- Emergency access if AAA server fails
- Privilege 15 gives full administrative rights

SSH CONFIGURATION

Step 8: Set Domain Name

```
ip domain-name lab.local
```

It forms part of router's identity, mandatory for RSA key generation. Without domain name, SSH cannot be enabled.

Step 9: Ensure Hostname is Set

Already set earlier:

```
hostname R1
```

- SSH uses hostname + domain name to create cryptographic keys.

Step 10: Generate RSA Keys

```
crypto key generate rsa
```

When prompted: How many bits? 1024

- Creates public/private key pair
- Enables encryption for SSH
- Activates SSH service on router

Step 11: Enable SSH Version 2

```
ip ssh version 2
```

Why: Version 2 is secure. Version 1 has vulnerabilities.

Step 12: Configure VTY Lines for AAA + SSH

```
line vty 0 4
transport input ssh
login authentication default
exit
```

Explanation:

- line vty 0 4 → Remote access lines
- transport input ssh → Only SSH allowed
- login authentication default → Use AAA server for login

Step 13: Save Configuration

```
end
write memory
```

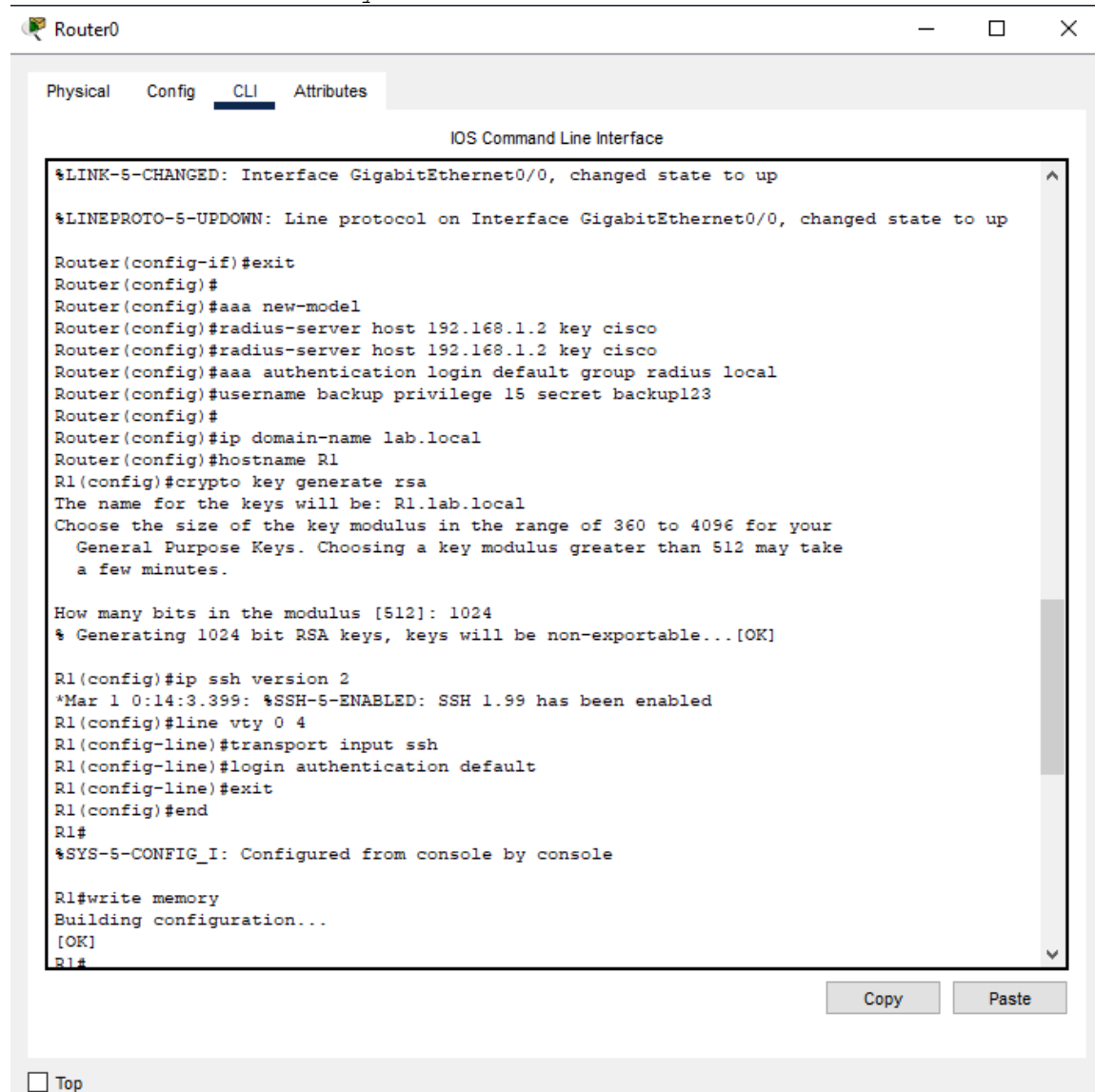


Fig.3. Steps 4 to 13

Step 14: Verify Network Connectivity

From PC Command Prompt:

```
ping 192.168.1.1
ping 192.168.1.2
```

Successful replies confirm connectivity. If ping fails → SSH will not work

Step 15: Test Remote Authentication

From PC:

```
ssh -l admin 192.168.1.1
```

Enter password “cisco123” when prompted.

What happens internally?

- PC requests SSH session
- Router asks AAA server to authenticate
- Server verifies username/password
- Access granted if valid

Step 16: Verify AAA Operation

On Router:

```
show ip ssh
show users
```

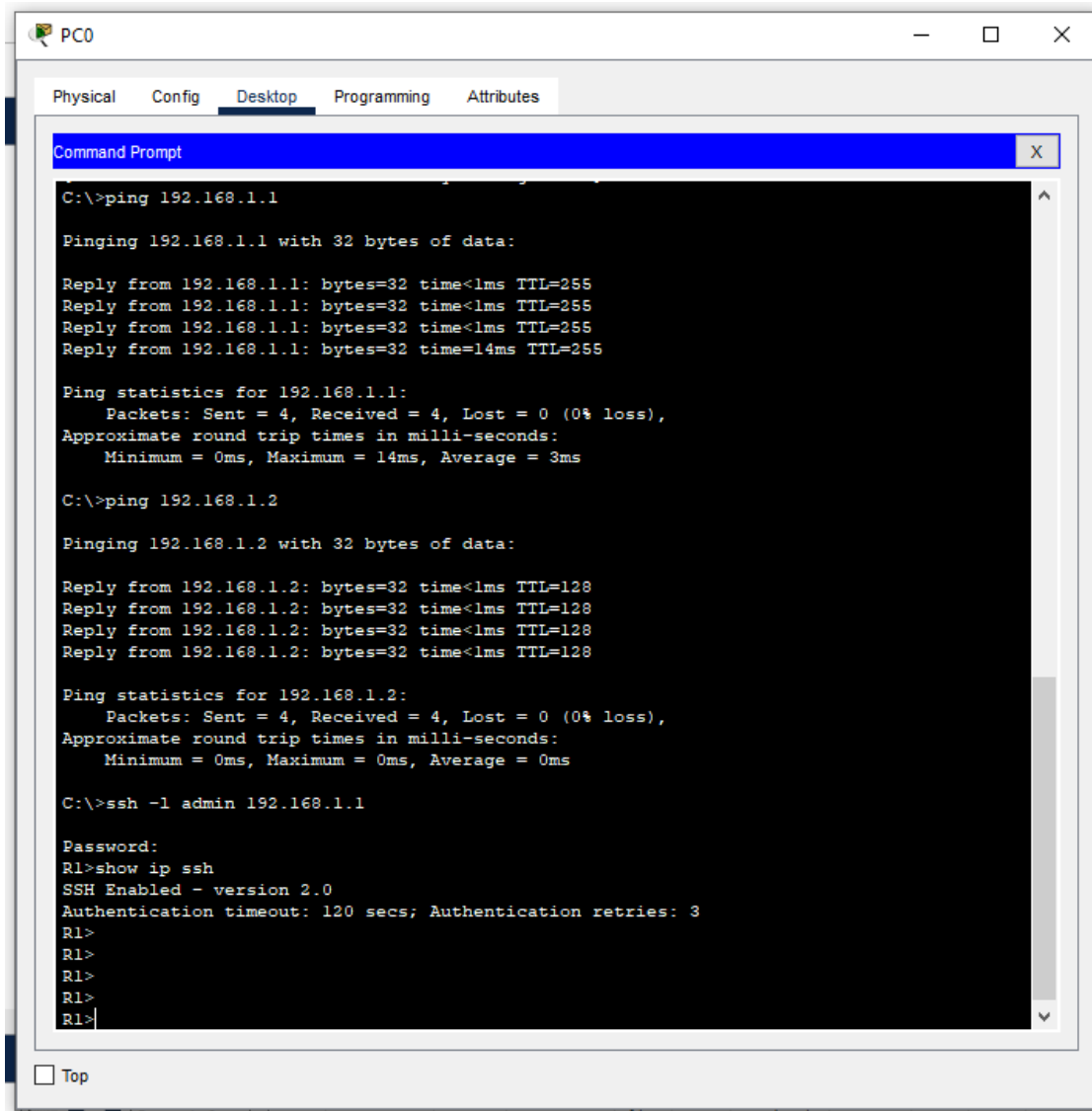


Fig.4. Steps 14 to 16